INCH-POUND
MIL-M-38510/130C
2 June 2004
SUPERSEDING
MIL-M-38510/130B
2 January 1986

MILITARY SPECIFICATION

MICROCIRCUITS, MONOLITHIC SILICON, BIPOLAR, INTERFACE, MEMORY CORE DRIVERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reactivated for new design as of 2 June 2004. May be used for either new or existing design acquisition.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-38535

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, bipolar, interface, memory core drivers. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3)

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Dual source and dual sink (positive – OR)
02	Quad sink (positive – OR)
03	Quad source/sink (positive – OR)

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outline. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, 3990 East Broad St., Columbus, OH 43218-3990, or emailed to bipolar@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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1.3 Absolute maximum ratings.

Supply voltage (V_{CC1})	7.0 V
Supply voltage (V_{CC2}) (device types 01, 03)	25 V
Input voltage (V_{IN})	5.5 V
Output collector voltage (V_{OCV}) (device types 02, 03)	25 V
Output clamp voltage (V_{OC}) (device type 02)	25 V
Output collector current (I_{OC})	750 mA
Duty cycle for each output transistor	20% 1/
Continuous total dissipation at T_A = 25°C	1375 mW
Storage temperature range	-65°C to +150°C
Junction temperature (T_J)	150°C
Lead temperature 1/16 inch from case, 60 seconds	300°C

1.4 Recommended operating conditions.

Supply voltage range	4.5 V to 5.5 V
Ambient operating temeperature range (T_A)	-55°C to +125°C

1.5 Power and thermal characteristics.

Case outline	Maximum allowable power dissipation 2/	Maximum θ_{JC}	Maximum θ_{JA}
E	275 mW @ T_A = 125°C	29°C/W	91°C/W
F	110 mW @ T_A = 125°C	70°C/W	220°C/W

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

1/ Warning: These devices do not pass the requirement for the glassivation layer integrity test (MIL-STD-883, method 2021) as specified in MIL-PRF-38535, A3.5.5.4. Therefore the glassivation layer integrity test is waived. Operation of these devices above the 20% duty cycle rating for each output transistor may result in device reliability problems.
2/ All leads welded or soldered to PC board.

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3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Logic diagrams and terminal connections. The circuit diagram and terminal connections shall be as specified on figure 1.

3.3.2 Truth table. The truth table shall be as specified on figure 2.

3.3.3 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity and the preparing activity (DSCC-VA) upon request.

3.3.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.8 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.9 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 55 (see MIL-PRF-38535, appendix A).

4. VERIFICATION.

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer’s Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Input clamp voltage	V_{IC}	V_{CC1} = 4.5 V, I_{IN} = -10 mA	01,02, 03	---	-1.7	V
Collectors terminal off-state current	I_{OFF}	V_{CC1} = 4.5 V, V_{COL} = 24 V	01,03	---	500	µA
High level input current	I_{IH1}	V_{CC1} = 5.5 V, V_{IN} = 2.4 V	01	---	40	µA
Address inputs	I_{IH2}	V_{CC1} = 5.5 V, V_{IN} = 5.5 V	01	---	1	mA
	I_{IH3}	V_{CC1} = 5.5 V, V_{IN} = 2.4 V	02,03	---	40	µA
	I_{IH4}	V_{CC1} = 5.5 V, V_{IN} = 5.5 V	02,03	---	1	mA
Strobe inputs	I_{IH5}	V_{CC1} = 5.5 V, V_{IN} = 2.4 V	01	---	80	µA
			02,03	---	160	
	I_{IH6}	V_{CC1} = 5.5 V, V_{IN} = 5.5 V	01	---	2	mA
			02,03	---	4	
Low level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	---	-3.2	mA
Address inputs	I_{IL2}		02,03	---	-6.4	
Strobe inputs	I_{IL3}		01,02, 03	---	-1.6	
Supply current from V_{CC1} All sources and sinks off	I_{CC1}	V_{CC1} = 5.5 V, V_{CC2} = 24 V	01	---	22	mA
	I_{CC2}	V_{CC1} = 5.5 V	02	---	25	
	I_{CC3}	V_{CC1} = 5.5 V, V_{CC2} = 24 V	03	---	10	
One sink on	I_{CC4}	V_{CC1} = 5.5 V, V_{CC2} = 24 V	01	---	70	mA
	I_{CC5}	V_{CC1} = 5.5 V	02	---	75	
One output on	I_{CC6}	V_{CC1} = 5.5 V, V_{COL} = 6 V, I_S = -50 mA	03	---	12	
Supply current from V_{CC2}, all sources and sinks off	I_{CC7}	V_{CC1} = 5.5 V, V_{CC2} = 24 V (pulse technique)	01,03	---	20	mA
One source on	I_{CC8}	V_{CC1} = 5.5 V, V_{CC2} = 24 V, I_S = -50 mA (pulse technique)	01	---	50	
One output on	I_{CC9}	V_{CC1} = 5.5 V, I_S = -50 mA, V_{CC2} = 6 V (pulse technique)	03	---	55	

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
Saturation voltage Source outputs	V_{CESAT1}	V_{CC1} = 4.5 V, V_{CC2} = 15 V, I_S = -600 mA (pulse technique)	01,03	---	0.9	V
Sink outputs	V_{CESAT2}		01,02	---	0.9	
High-level sink output voltage	V_{OH}	V_{CC1} = 4.5 V, V_{CC2} = 24 V, I_O = 0	01,02	19	---	V
Output clamp diode forward voltage	V_{OC}	V_C = 0, I_C = -10 mA	02	---	1.5	V
Output clamp current, one output on	I_{OC}	I_S = 50 mA	02	---	7	mA
Propagation delay time (low-to-high level output) Source collectors	t_{PLH1}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω, T_A = 25°C	01,03	---	50	ns
Sink outputs	t_{PLH2}		01,02	---	45	
Propagation delay time (high-to-low level output) Source collectors	t_{PHL1}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω, T_A = 25°C	01,03	---	50	ns
Sink outputs	t_{PHL2}		01,02	---	30	
Transition time (low-to-high level output) Source outputs	t_{TLH1}	V_{CC1} = 5 V, V_{CC2} = 20 V, C_L = 100 pF, R_L = 100 kΩ, T_A = 25°C	01	---	30	ns
	t_{TLH2}	V_{CC1} = 5 V, V_{CC2} = 20 V, C_L = 100 pF, R_L = 100 Ω, T_A = 25°C	03	---	15	
Sink outputs	t_{TLH3}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω, T_A = 25°C	01,02	---	15	
Transition time (high-to-low-level output) Source outputs	t_{THL1}	V_{CC1} = 5 V, V_{CC2} = 20 V, C_L = 100 pF, R_L = 100 kΩ, T_A = 25°C	01	---	25	ns
	t_{THL2}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 100 Ω, T_A = 25°C	03	---	18	
Sink outputs	t_{THL3}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω, T_A = 25°C	01,02	---	15	
Storage time, sink outputs	t_S	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω, T_A = 25°C	01,02	---	35	ns

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ unless otherwise specified	Device types	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC1} = 5 V, $V_S = V_{CLAMP}$ = 24 V, C_L = 100 pF, R_L = 470 Ω, T_A = 25°C	02,03	V_S – 0.25		V
Propagation delay time (low-to-high level output) Source collectors	t_{PLH1}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω	01,03	---	90	ns
Sink outputs	t_{PLH2}		01,02	---	65	
Propagation delay time (high-to-low level output) Source collectors	t_{PHL1}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω	01,03	---	65	ns
Sink outputs	t_{PHL2}		01,02	---	45	
Transition time (low-to-high level output) Source outputs	t_{TLH1}	V_{CC1} = 5 V, V_{CC2} = 20 V, C_L = 100 pF, R_L = 100 kΩ	01	---	45	ns
	t_{TLH2}	V_{CC1} = 5 V, V_{CC2} = 20 V, C_L = 100 pF, R_L = 100 Ω	03	---	20	
Sink outputs	t_{TLH3}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω	01,02	---	20	
Transition time (high-to-low-level output) Source outputs	t_{THL1}	V_{CC1} = 5 V, V_{CC2} = 20 V, C_L = 100 pF, R_L = 100 kΩ	01	---	27	ns
	t_{THL2}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 100 Ω	03	---	20	
Sink outputs	t_{THL3}	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω	01,02	---	20	
Storage time, sink outputs	t_S	V_{CC1} = 5 V, V_{CC2} = 15 V, C_L = 100 pF, R_L = 24 Ω	01,02	---	55	ns
High-level output voltage	V_{OH}	V_{CC1} = 5 V, $V_S = V_{CLAMP}$ = 24 V, C_L = 100 pF, R_L = 470 Ω	02,03	V_S – 0.25		V

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TABLE II. Electrical test requirements.

MIL-PRF-38535 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters	1	1
Final electrical test parameters	1*, 2, 3, 9	1*, 2, 3, 9
Group A test requirements	1,2,3,9,10,11	1,2,3,9,10,11
Group B electrical test parameters when using the method 5005 QCI option	1,2,3 and table IV delta limits	N/A
Group C end-point electrical parameters	1,2,3 and table IV delta limits	See 4.4.3
Additional electrical subgroups for group C periodic inspections	N/A	N/A
Group D end-point electrical parameters	1,2,3,	1,2,3

* PDA applies to subgroup 1.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

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Device type 01

Cases E and F

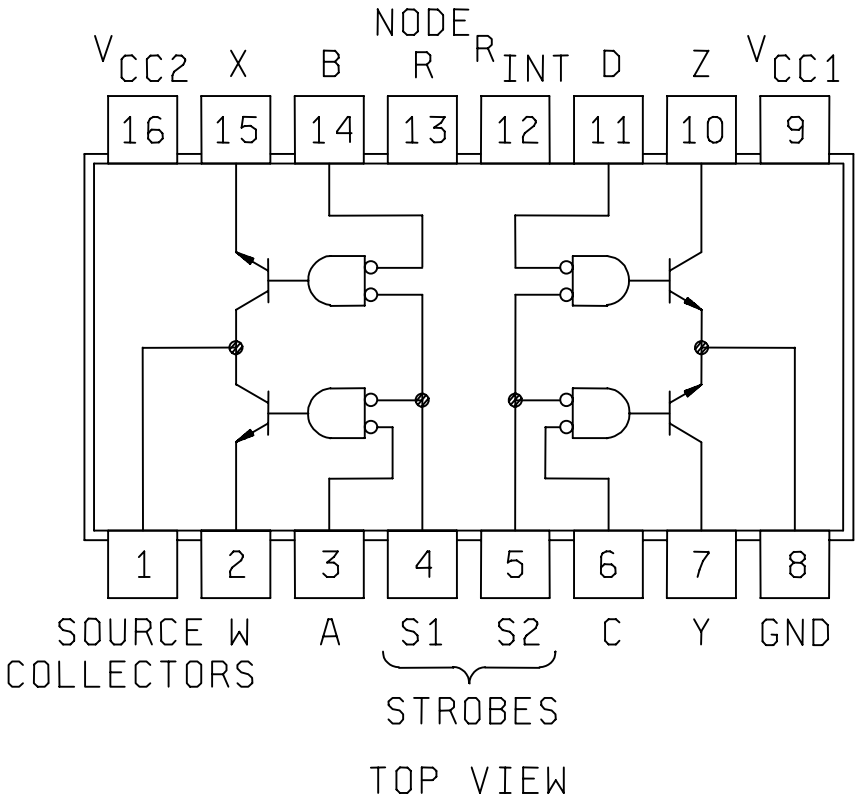


Figure 1. Logic diagram and terminal connections.

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Device type 02

Cases E and F

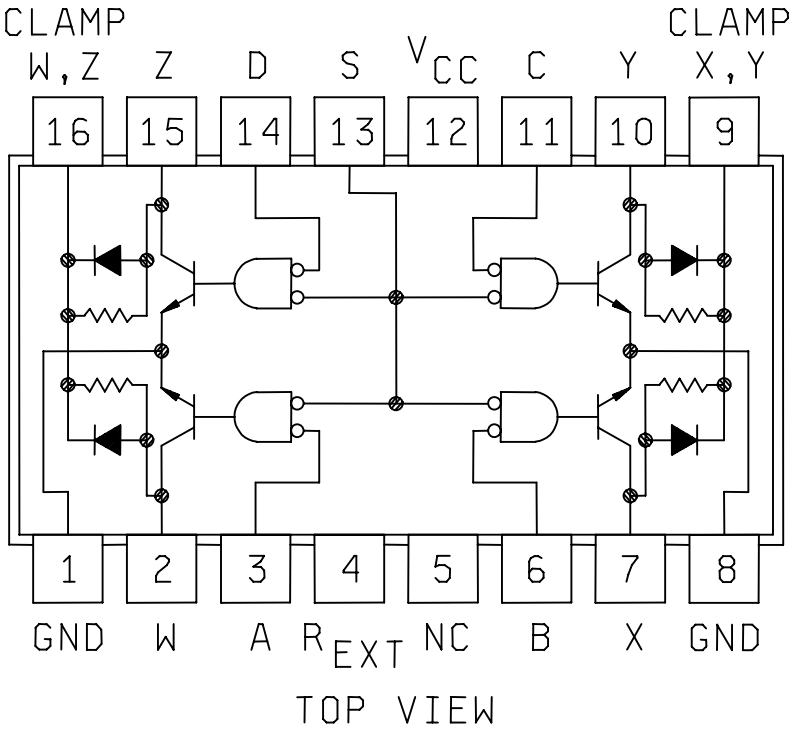


Figure 1. Logic diagram and terminal connections – Continued.

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Device type 03

Case E

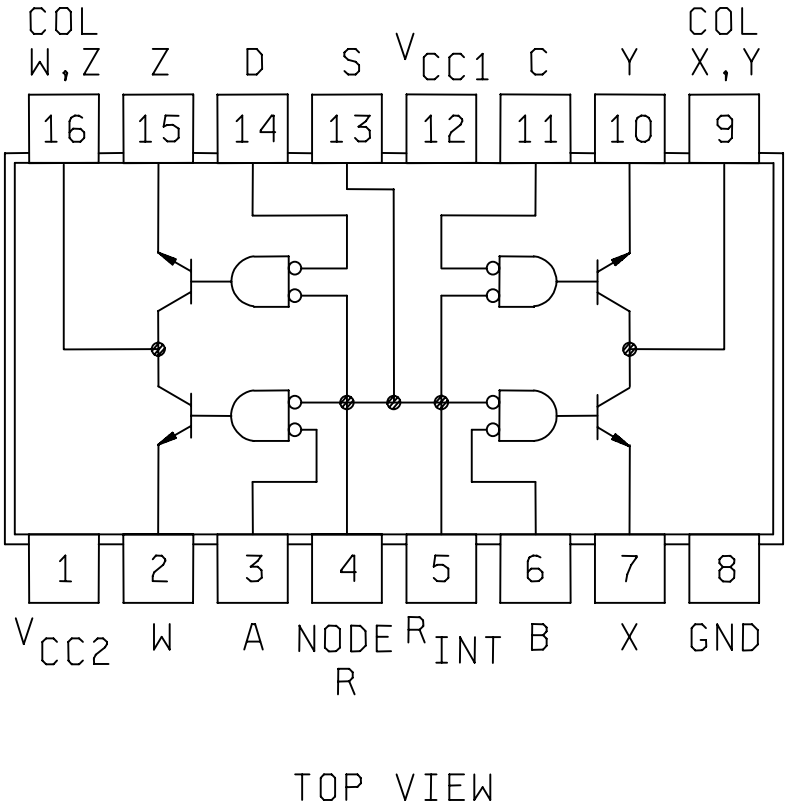


Figure 1. Logic diagram and terminal connections – Continued.

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Device type 01

Address Inputs				Strobe Inputs		Outputs			
Source		Sink		Source	Sink	Source		Sink	
A	B	C	D	S1	S2	W	X	Y	Z
L	H	X	X	L	H	ON	OFF	OFF	OFF
H	L	X	X	L	H	OFF	ON	OFF	OFF
X	X	L	H	H	L	OFF	OFF	ON	OFF
X	X	H	L	H	L	OFF	OFF	OFF	ON
X	X	X	X	H	H	OFF	OFF	OFF	OFF
H	H	H	H	X	X	OFF	OFF	OFF	OFF

H = High level, L = Low level, X = Irrelevant
Note: Not more than one output is to be on at any one time.

Device type 02

Inputs					Outputs			
Address				Strobe	W	X	Y	Z
A	B	C	D	S				
L	H	H	H	L	ON	OFF	OFF	OFF
H	L	H	H	L	OFF	ON	OFF	OFF
H	H	L	H	L	OFF	OFF	ON	OFF
H	H	H	L	L	OFF	OFF	OFF	ON
H	H	H	H	X	OFF	OFF	OFF	OFF
X	X	X	X	H	OFF	OFF	OFF	OFF

H = High level, L = Low level, X = Irrelevant
Note: Not more than one output is to be on at any one time.

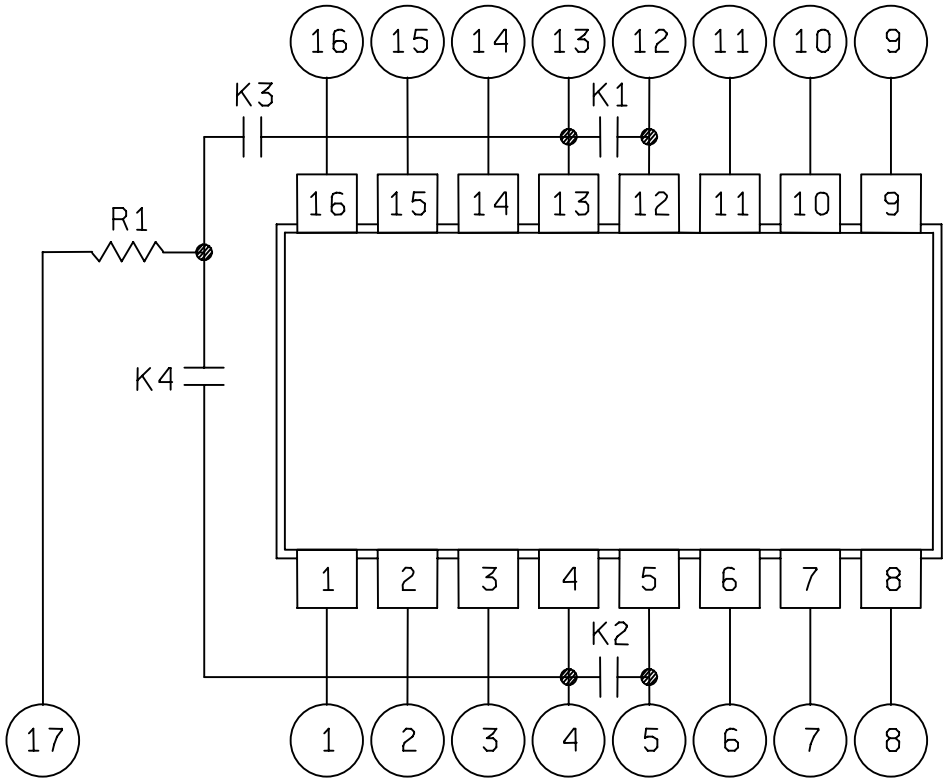
Device type 03

Inputs					Outputs			
Address				Strobe	W	X	Y	Z
A	B	C	D	S				
L	H	H	H	L	ON	OFF	OFF	OFF
H	L	H	H	L	OFF	ON	OFF	OFF
H	H	L	H	L	OFF	OFF	ON	OFF
H	H	H	L	L	OFF	OFF	OFF	ON
H	H	H	H	X	OFF	OFF	OFF	OFF
X	X	X	X	H	OFF	OFF	OFF	OFF

H = High level, L = Low level, X = Irrelevant
Note: Not more than one output is to be on at any one time.

Figure 2. Truth table.

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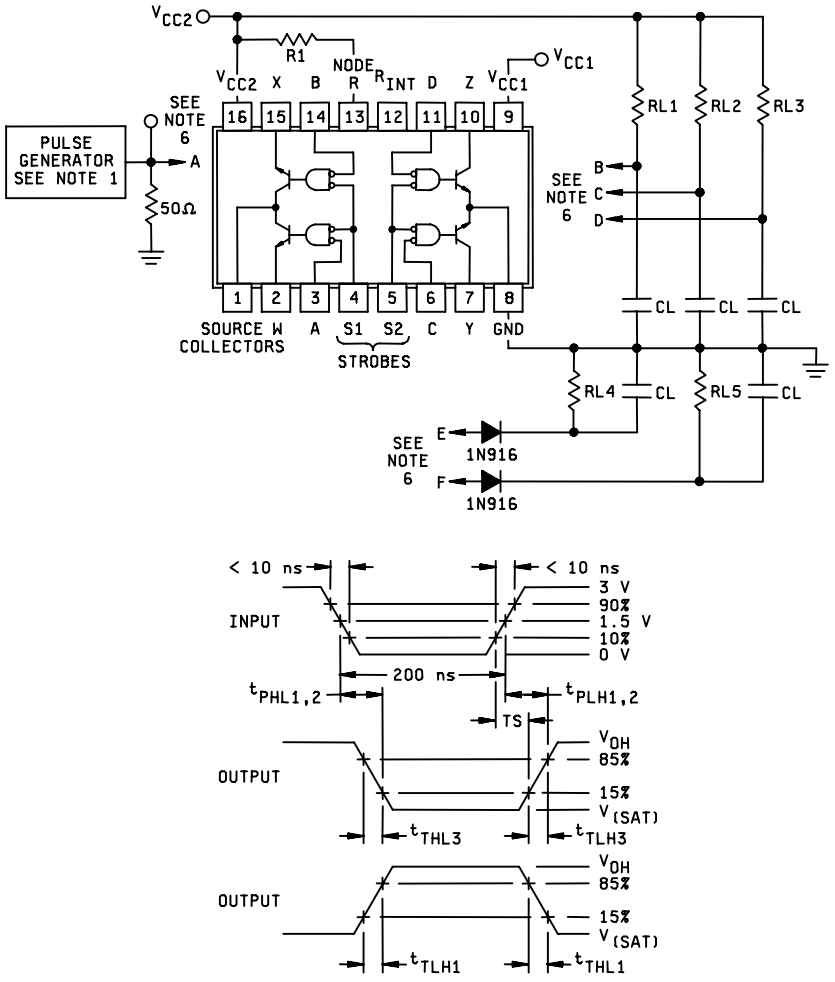


NOTES:
1. All relays are in the de-energized position.
2. R1 = 360 Ω ±5% carbon.

FIGURE 3. Static test circuit.

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Device type 01

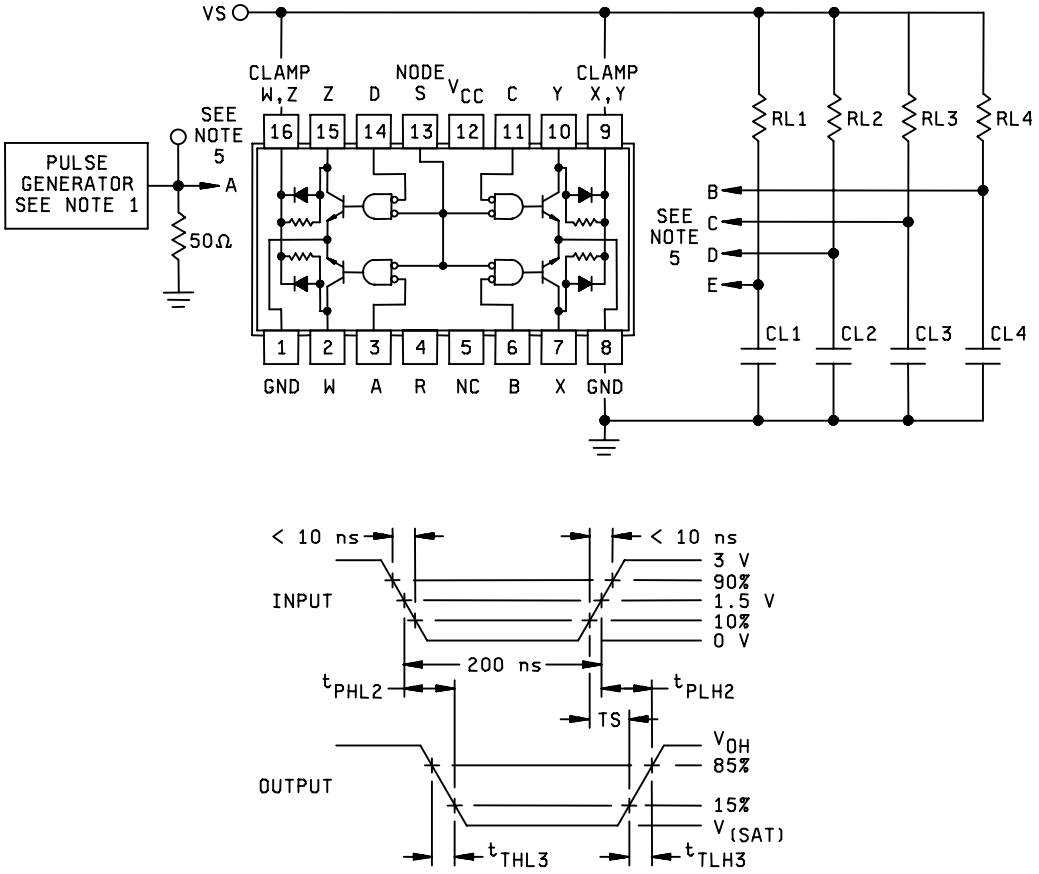


NOTES:
1. The pulse generator has the following characteristics:
   TP = 200 ns, PRR ≤ 10 Hz, TR = TF ≤ 10 ns, Z_{OUT} = 50 Ω.
2. R1 = 360 Ω ±5% carbon.
3. RL1-3 = 24 Ω ±5% carbon.
4. RL4,5 = 100 Ω ±5% carbon.
5. CL = 100 pF ±5%, including probe and jig capacitance.
6. Connect loads and pulse generator to terminals as specified in table III.

FIGURE 4. Switching time.

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Device type 02

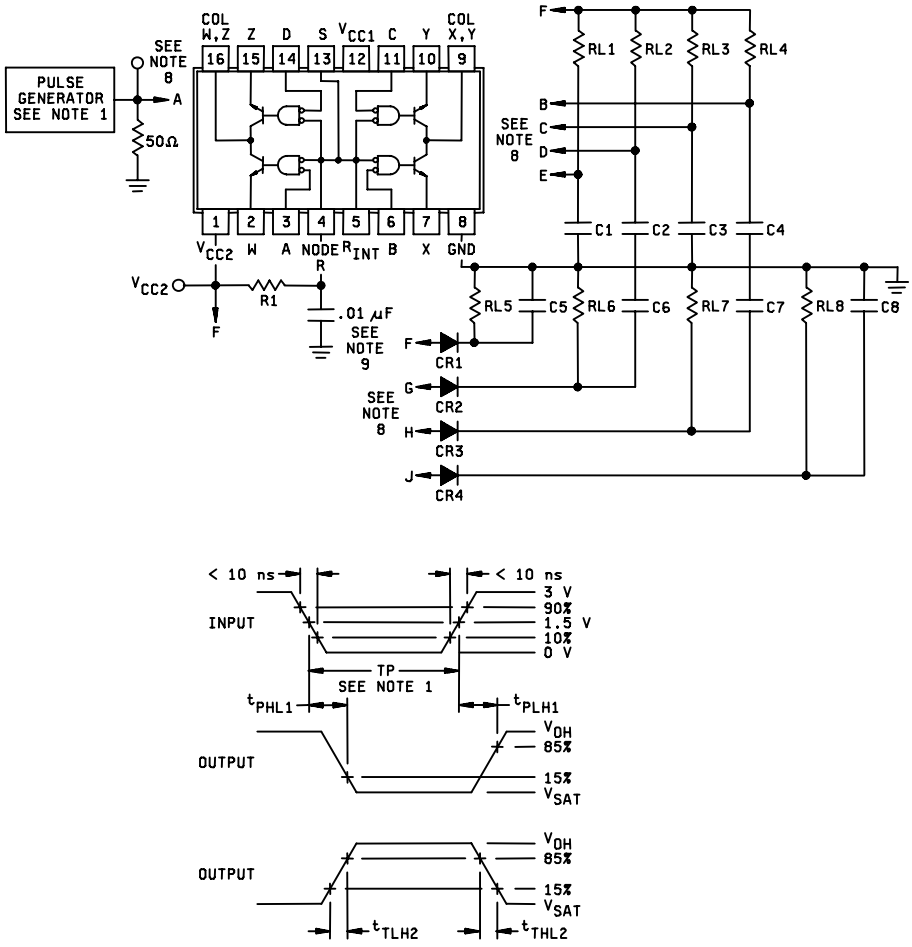


NOTES:
1. The pulse generator has the following characteristics:
   TP = 200 ns, PRR ≤ 10 Hz, TR = TF ≤ 10 ns, Z_{OUT} = 50 Ω.
   For all tests except V_{OH} TP = 40 µs.
2. RL1 = RL4 = 470 Ω ±5% carbon for V_{OH} and 24 Ω ±5% carbon for all other tests.
3. CL1-CL4 = 100 pF ±5%, including probe and jig capacitance.
4. Connect VS to V(Clamp)
5. Connect loads and pulse generator to terminals as specified in table III.

FIGURE 4. Switching time – Continued.

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Device type 03



NOTES:
1. The pulse generator has the following characteristics:
   TP = 200 ns, PRR ≤ 10 Hz, TR = TF ≤ 10 ns, Z_{OUT} = 50 Ω.
2. R1 = 360 Ω ±5% carbon.
3. RL1-2 = 24 Ω ±5% carbon.
4. RL3-4 = 470 Ω ±5% carbon.
5. RL5-8 = 100 Ω ±5% carbon.
6. C1-8 = 100 pF ±5% (including probe and jig capacitance).
7. CR1-4 = 1N916 or equivalent.
8. Connect loading and pulse generator to terminals as specified in table III.
9. For use only during the t_{TLH2} and t_{THL2} tests.

FIGURE 4. Switching time – Continued.

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TABLE III. Group A inspection.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
1 T_A = +25°C	V_{IC}	1		GND	-10 mA	4.5 V	4.5 V	4.5 V		GND	4.5 V		4.5 V			4.5 V	GND	24 V		K1	3		3		-1.7	V
		2		"	4.5 V	"	"	-10 mA		"	"		4.5 V			"	"	"		"	6		"		"	"
		3		"	"	"	"	4.5 V		"	"		-10 mA			"	"	"		"	11	01	"		"	"
		4		"	"	"	"	"		"	"		4.5 V			-10 mA	"	"		"	14		"		"	"
		5		"	"	-10 mA	"	"		"	"		"			4.5 V	"	"		"	4		"		"	"
		6		"	"	4.5 V	-10 mA	"		"	"		"			"	"	"		"	5		"		"	"
		7	GND		-10 mA	"		"		"	24 V		"	4.5 V	4.5 V	"					3		"		"	"
		8	"		4.5 V	"		-10 mA		"	"		"	"	"	"					6	02	"		"	"
		9	"		"	"		"		"	"		-10 mA	"	-10 mA	-10 mA					11		"		"	"
		10	"		"	"		"		"	"		4.5 V	"	"	4.5 V					14		"		"	"
		11	"		-10 mA	"		"		"	"		"	"	4.5 V	"					13		"		"	"
		12	24 V		4.5 V	"		"		"	"		"	"	"	"					3		"		"	"
		13	"		"	"		-10 mA		"	"		"	"	"	"					6	03	"		"	"
		14	"		"	"		4.5 V		"	"		-10 mA	"	"	-10 mA					11		"		"	"
		15	"		"	"		"		"	"		4.5 V	"	-10 mA	4.5 V					14		"		"	"
		16	"		"	"		"		"	"		4.5 V	"	"	"					13		"		"	"
	I_{OFF}	17	24 V	GND	GND	2 V	4.5 V	4.5 V		"	4.5 V					GND	GND	"		K1	1	01	"		500	µA
		18	"	"	2 V	GND	4.5 V	"		"	4.5 V					2 V	"	"		K1	1		"		"	"
		19	"	"	GND				GND	"	24 V	GND		4.5 V	2 V	GND	"	"		K2	16	03	"		"	"
		20	"	"	2 V				"	"	"	"		"	GND	2 V	"	"		"	16		"		"	"
		21	"	"				GND	"	"	"	"	GND	"	2 V		"	"		"	9		"		"	"
		22	"	"				2 V	"	"	"	"	2 V	"	GND		"	"		"	9		"		"	"
	V_{OH}	23	"	"	4.5 V	4.5 V	GND	2 V		"	4.5 V		4.5 V			4.5 V	"	"		K1	7	01	"	19		V
		24	"	"	"	"	2 V	GND		"	"		4.5 V			"	"	"		"	7		"	"		"
		25	"	"	"	"	GND	4.5 V		"	"		2 V			"	"	"		"	10		"	"		"
		26	"	"	"	"	2 V	4.5 V		"	"		GND			"	"	"		"	10		"	"		"
		27	GND		2 V					"	24 V			4.5 V	GND	"					2		"	"		"
		28	"		GND					"	"			"	2 V	"					2		"	"		"
		29	"		4.5 V					"	"			"	GND	2 V					15		"	"		"
		30	"		4.5 V					"	"			"	2 V	GND					15	02	"	"		"
		31	"					2 V		"	"		4.5 V	"	GND						7		"	"		"
		32	"					GND		"	"		4.5 V	"	2 V						7		"	"		"
		33	"					4.5 V		"	"		2 V	"	GND						10		"	"		"
		34	"					4.5 V		"	"		GND	"	2 V						10		"	"		"
	V_{CESAT1}	35	600 mA	GND	0.8 V	0.8 V				"	4.5 V					4.5 V	GND	15 V	15 V	K3	1	01	"		0.9	"
		36	600 mA	"	4.5 V	0.8 V				"	4.5 V					0.8 V	GND	15 V	15 V	K3	1		"		"	"
		37	24 V	GND				4.5 V	GND	"	600 mA	GND	0.8 V	4.5 V	0.8 V		"	24 V	24 V	K4	9		"		"	"
		38	"	"				0.8 V	"	"	600 mA	"	4.5 V	"	"		"	24 V	"	K4	9	03	"		"	"
		39	"	"	4.5 V				"	"	24 V	"		"	"	0.8 V	"	600 mA	"	K4	16		"		"	"
		40	"	"	0.8 V				"	"	24 V	"		"	"	4.5 V	"	600 mA	"	K4	16		"		"	"
	V_{CESAT2}	41					0.8 V	4.5 V		"	4.5 V	600 mA	0.8 V					15 V	15 V	K1	10	01	"		"	"
		42					0.8 V	0.8 V	600 mA	"	4.5 V		4.5 V					"	15 V	K1	7		"		"	"
		43	GND	600 mA	0.8 V					"	15 V			4.5 V	0.8 V	4.5 V		"	24 V		2		"		"	"
		44	"					0.8 V	600 mA	"	"			"	"			"	"		7	02	"		"	"
		45	"					4.5 V		"	"	600 mA	4.5 V	"	"			"	"		10		"		"	"
		46	"		4.5 V					"	"		0.8 V	"	"	0.8 V	600 mA	"	"		15		"		"	"
	V_{OC}	47	"	10 mA	"			4.5 V		"	GND		4.5 V	5.5 V	4.5 V	4.5 V		GND			2		"		1.5	"
		48	"		"			"	10 mA	"	"		"	"	"	"		"			7	02	"		"	"
		49	"		"			"		"	"	10 mA	"	"	"	"		"			10		"		"	"
		50	"		"			"		"	"		"	"	"	"	10 mA	"			15		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
1 T_A = 25°C	I_{OC}	51	GND		4.5 V					GND				5.5 V	0.8 V	0.8 V	50 mA	24 V			16		3		7	mA
		52	"	50 mA	0.8 V					"				"	"	4.5 V		24 V			16	02	"		"	"
		53	"					4.5 V		"	24 V	50 mA	0.8 V	"	"						9		"		"	"
		54	"					0.8 V	50 mA	"	24 V		4.5 V	"	"						9		"		"	"
	I_{IH1}	55			2.4 V	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			5.5 V		24 V		K1	3		"		40	µA
		56			5.5 V	GND	5.5 V	5.5 V		"	"		"			2.4 V		"		"	14	01	"		"	"
		57			"	5.5 V	GND	2.4 V		"	"		"			5.5 V		"		"	6		"		"	"
		58			"	5.5 V	GND	5.5 V		"	"		2.4 V			"		"		"	11		"		"	"
	I_{IH2}	59			5.5 V	GND	5.5 V	"		"	"		5.5 V			"		"		"	3		"		1	mA
		60			"	GND	5.5 V	"		"	"		"			"		"		"	14	01	"		"	"
		61			"	5.5 V	GND	"		"	"		"			"		"		"	6		"		"	"
		62			"	5.5 V	GND	"		"	"		"			"		"		"	11		"		"	"
	I_{IH3}	63	GND		2.4 V			"		"			"	5.5 V	GND	"					3		"		40	µA
		64	"		5.5 V			2.4 V		"			"	"	"	"					6	02	"		"	"
		65	"		5.5 V			5.5 V		"			2.4 V	"	"	"					11		"		"	"
		66	"		5.5 V			5.5 V		"			5.5 V	"	"	2.4 V					14		"		"	"
		67	24 V		2.4 V			5.5 V		"			"	"	"	5.5 V				K2	3		"		"	"
		68	"		5.5 V			2.4 V		"			"	"	"	"				"	6	03	"		"	"
		69	"		"			5.5 V		"			2.4 V	"	"	"				"	11		"		"	"
		70	"		"			"		"			5.5 V	"	"	2.4 V				"	14		"		"	"
	I_{IH4}	71	GND		5.5 V			"		"			"	"	"	5.5 V					3		"		1	mA
		72	"		"			"		"			"	"	"	"					6	02	"		"	"
		73	"		"			"		"			"	"	"	"					11		"		"	"
		74	"		"			"		"			"	"	"	"					14		"		"	"
		75	24 V		"			"		"			"	"	"	"				K2	3		"		"	"
		76	"		"			"		"			"	"	"	"				"	6	03	"		"	"
		77	"		"			"		"			"	"	"	"				"	11		"		"	"
		78	"		"			"		"			"	"	"	"				"	14		"		"	"
	I_{IH5}	79			GND	2.4 V	5.5 V	"		"	5.5 V		"			GND		24 V		K1	4	01	"		80	µA
		80			5.5 V	5.5 V	2.4 V	GND		"	"		GND			5.5 V		24 V		K1	5		"		80	"
		81	GND		GND			"		"			"	5.5 V	2.4 V	GND					13	02	"		160	"
		82	24 V		"			"		"			"	5.5 V	2.4 V	"				K2	13	03	"		160	"
	I_{IH6}	83			"	5.5 V	5.5 V	5.5 V		"	5.5 V		5.5 V			"		24 V		K1	4	01	"		2	mA
		84			5.5 V	5.5 V	5.5 V	GND		"	5.5 V		GND			5.5 V		24 V		K1	5		"		2	"
		85	GND		GND			"		"			"	5.5 V	5.5 V	GND					13	02	"		4	"
		86	24 V		GND			"		"			"	5.5 V	5.5 V	GND				K2	13	03	"		4	"
	I_{IL1}	87			5.5 V	0.4 V	5.5 V	5.5 V		"	5.5 V		5.5 V			5.5 V		24 V		K1	4	01	"		-3.2	"
		88			"	5.5 V	0.4 V	"		"	5.5 V		"			"		24 V		K1	5		"		-3.2	"
	I_{IL2}	89	GND		"			"		"			"	5.5 V	0.4 V	"					13	02	"		-6.4	"
		90	24 V		"			"		"			"	5.5 V	0.4 V	"				K2	13	03	"		-6.4	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open)																	Relays energized	Measured terminal	Device	Test Figure	Limits		Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17					Min	Max	
1 T_A = 25°C	I_{IL3}	91			0.4 V	5.5 V	5.5 V	5.5 V		GND	5.5 V		5.5 V			5.5 V		24 V		K1	3		3		-1.6	mA
		92			5.5 V	"	"	5.5 V		"	"		"			0.4 V		"		"	14	01	"		"	"
		93			"	"	"	0.4 V		"	"		"			5.5 V		"		"	6		"		"	"
		94			"	"	"	5.5 V		"	"		0.4 V			"		"		"	11		"		"	"
		95	GND		0.4 V			"		"			5.5 V	5.5 V	5.5 V	"					3		"		"	"
		96	"		5.5 V			"		"			"	"	"	0.4 V					14	02	"		"	"
		97	"		"			0.4 V		"			"	"	"	5.5 V					6		"		"	"
		98	"		"			5.5 V		"			0.4 V	"	"	"					11		"		"	"
		99	24 V		0.4 V			"		"			5.5 V	"	"	"				K2	3		"		"	"
		100	"		5.5 V			"		"			"	"	"	0.4 V				"	14	03	"		"	"
		101	"		"			0.4 V		"			"	"	"	5.5 V				"	6		"		"	"
		102	"		"			5.5 V		"			0.4 V	"	"	5.5 V				"	11		"		"	"
	I_{CC1}	103		GND	5 V	5 V	5 V	5 V		"	5.5 V		5 V			5 V	GND	24 V		K1	9	01	"		22	"
	I_{CC2}	104	GND		5 V			5 V		"			"	5.5 V	5 V	5 V					12	02	"		25	"
	I_{CC3}	105	24 V	GND	5 V			5 V	GND	"		GND	"	5.5 V	5 V	5 V	GND			K2	12	03	"		10	"
	I_{CC4}	106		"	4.5 V	4.5 V	GND	GND	50 mA	"	5.5 V		"			4.5 V	GND	24 V		K1	9	01	"		70	"
		107		"	4.5 V	4.5 V	GND	5 V		"	5.5 V	50 mA	GND			"	"	"			9		"		70	"
	I_{CC5}	108	GND	50 mA	GND			5 V		"	24 V		5 V	5.5 V	GND	5 V		"			12		"		75	"
		109	"		5 V			GND	50 mA	"	"		5 V	"	"	"		"			"	02	"		"	"
		110	"		"			5 V		"	"	50 mA	GND	"	"	"		"			"		"		"	"
		111	"		"			"		"	"		5 V	"	"	GND	50 mA	"			"		"		"	"
	I_{CC6}	112	24 V	-50 mA	GND			"		"	6 V		"	"	"	5 V		6 V		K2	"		"		12	"
		113	"		5 V			GND	-50 mA	"	"		"	"	"	"		"		"	"	03	"		"	"
		114	"		"			5 V		"	"	-50 mA	GND	"	"	"		"		"	"		"		"	"
		115	"		"			"		"	"		5 V	"	"	GND	-50 mA	"		"	"		"		"	"
	I_{CC7}	116		GND	"	5 V	5 V	"		"	5.5 V		5 V			5 V	GND	24 V		K1	16	01	"		20	"
		117	24 V	GND	"			"	GND	"		GND	5 V	5.5 V	5 V	5 V	GND			K2	1	03	"		20	"
	I_{CC8}	118	50 mA	GND	GND	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			5 V	GND	24 V		K1	16	01	"		50	"
		119	50 mA	GND	5 V	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			GND	GND	24 V		K1	16		"		50	"
	I_{CC9}	120	24 V	-50 mA	GND			5 V		"	6 V		5 V	5.5 V	GND	5 V		6 V		K2	1		"		55	"
		121	"		5 V			GND	-50 mA	"	"		5 V	"	"	"		"		"	"	03	"		"	"
		122	"		"			5 V		"	"	-50 mA	GND	"	"	"		"		"	"		"		"	"
		123	"		"			5 V		"	"		5 V	"	"	GND	-50 mA	"		"	"		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
2 T_A = 125°C	V_{IC}	124		GND	-10 mA	4.5 V	4.5 V	4.5 V		GND	4.5 V		4.5 V			4.5 V	GND	24 V		K1	3		3		-1.7	V
		125		"	4.5 V	"	"	-10 mA		"	"		4.5 V			"	"	"		"	6		"		"	"
		126		"	"	"	"	4.5 V		"	"		-10 mA			"	"	"		"	11	01	"		"	"
		127		"	"	"	"	"		"	"		4.5 V			-10 mA	"	"		"	14		"		"	"
		128		"	"	-10 mA	"	"		"	"		"			4.5 V	"	"		"	4		"		"	"
		129		"	"	4.5 V	-10 mA	"		"	24 V		"			"	"	"		"	5		"		"	"
		130	GND		-10 mA	"		"		"	"		"	4.5 V	4.5 V	"		"			3		"		"	"
		131	"		4.5 V	"		-10 mA		"	"		"	"	"	"		"			6	02	"		"	"
		132	"		"	"		4.5 V		"	"		-10 mA	"	"	-10 mA		"			11		"		"	"
		133	"		"	"		"		"	"		4.5 V	"	"	4.5 V		"			14		"		"	"
		134	"		"	"		"		"	"		"	"	-10 mA	"		"			13		"		"	"
		135	24 V		-10 mA	"		"		"	"		"	"	4.5 V	"		"			3		"		"	"
		136	"		4.5 V	"		-10 mA		"	"		"	"	"	"		"			6	03	"		"	"
		137	"		"	"		4.5 V		"	"		-10 mA	"	"	"		"			11		"		"	"
		138	"		"	"		"		"	"		4.5 V	"	"	-10 mA		"			14		"		"	"
		139	"		"	"		"		"	"		4.5 V	"	-10 mA	4.5 V		"			13		"		"	"
	I_{OFF}	140	"	GND	GND	2 V	4.5 V	4.5 V		"	4.5 V					GND	GND	"		K1	1	01	"		500	µA
		141	"	"	2 V	GND	4.5 V	4.5 V		"	4.5 V					2 V	"	"		K1	1		"		"	"
		142	"	"	GND				GND	"	24 V	GND		4.5 V	2 V	GND	"	"		K2	16		"		"	"
		143	"	"	2 V				"	"	"	"		"	GND	2 V	"	"		"	16	03	"		"	"
		144	"	"				GND	"	"	"	"	GND	"	2 V		"	"		"	9		"		"	"
		145	"	"				2 V	"	"	"	"	2 V	"	GND		"	"		"	9		"		"	"
	V_{OH}	146	"	"	4.5 V	4.5 V	GND	2 V		"	4.5 V		4.5 V			4.5 V	"	"		K1	7	01	"	19		V
		147	"	"	"	"	2 V	GND		"	"		4.5 V			"	"	"		"	7		"	"		"
		148	"	"	"	"	GND	4.5 V		"	"		2 V			"	"	"		"	10		"	"		"
		149	"	"	"	"	2 V	4.5 V		"	"		GND			"	"	"		"	10		"	"		"
		150	GND		2 V					"	24 V			4.5 V	GND	"					2		"	"		"
		151	"		GND					"	"			"	2 V	"					2		"	"		"
		152	"		4.5 V					"	"			"	GND	2 V					15		"	"		"
		153	"		4.5 V					"	"			"	2 V	GND					15	02	"	"		"
		154	"					2 V		"	"		4.5 V	"	GND						7		"	"		"
		155	"					GND		"	"		4.5 V	"	2 V						7		"	"		"
		156	"					4.5 V		"	"		2 V	"	GND						10		"	"		"
		157	"					4.5 V		"	"		GND	"	2 V						10		"	"		"
	V_{CESAT1}	158	600 mA	GND	0.8 V	0.8 V				"	4.5 V					4.5 V	GND	15 V	15 V	K3	1	01	"		0.9	"
		159	600 mA	"	4.5 V	0.8 V				"	4.5 V					0.8 V	GND	15 V	15 V	K3	1		"		"	"
		160	24 V	GND				4.5 V	GND	"	600 mA	GND	0.8 V	4.5 V	0.8 V		"	24 V	24 V	K4	9	03	"		"	"
		161	"	"				0.8 V	"	"	600 mA	"	4.5 V	"	"		"	24 V	"	K4	9		"		"	"
		162	"	"	4.5 V				"	"	24 V	"		"	"	0.8 V	"	600 mA	"	K4	16		"		"	"
		163	"	"	0.8 V				"	"	24 V	"		"	"	4.5 V	"	600 mA	"	K4	16		"		"	"
	V_{CESAT2}	164					0.8 V	4.5 V		"	4.5 V	600 mA	0.8 V					15 V	15 V	K1	10	01	"		"	"
		165					0.8 V	0.8 V	600 mA	"	4.5 V		4.5 V					"	15 V	K1	7		"		"	"
		166	GND	600 mA	0.8 V					"	15 V			4.5 V	0.8 V	4.5 V		"	24 V		2		"		"	"
		167	"					0.8 V	600 mA	"	"			"	"			"	"		7	02	"		"	"
		168	"					4.5 V		"	"	600 mA	4.5 V	"	"			"	"		10		"		"	"
		169	"		4.5 V					"	"		0.8 V	"	"	0.8 V	600 mA	"	"		15		"		"	"
	V_{OC}	170	"	10 mA	"			4.5 V		"	GND		4.5 V	5.5 V	4.5 V	4.5 V			GND		2		"		1.5	"
		171	"		"			"	10 mA	"	"		"	"	"	"			"		7	02	"		"	"
		172	"		"			"		"	"	10 mA	"	"	"	"			"		10		"		"	"
		173	"		"			"		"	"		"	"	"	"	10 mA		"		15		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open)																	Relays energized	Measured terminal	Device	Test Figure	Limits		Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17					Min	Max	
2 T_A = 125°C	I_{OC}	174	GND		4.5 V					GND				5.5 V	0.8 V	0.8 V	50 mA	24 V			16		3		7	mA
		175	"	50 mA	0.8 V					"				"	"	4.5 V		24 V			16	02	"		"	"
		176	"					4.5 V		"	24 V	50 mA	0.8 V	"	"						9		"		"	"
		177	"					0.8 V	50 mA	"	24 V		4.5 V	"	"						9		"		"	"
	I_{IH1}	178			2.4 V	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			5.5 V		24 V		K1	3		"		40	µA
		179			5.5 V	GND	5.5 V	5.5 V		"	"		"			2.4 V		"		"	14	01	"		"	"
		180			"	5.5 V	GND	2.4 V		"	"		"			5.5 V		"		"	6		"		"	"
		181			"	5.5 V	GND	5.5 V		"	"		2.4 V			"		"		"	11		"		"	"
	I_{IH2}	182			5.5 V	GND	5.5 V	"		"	"		5.5 V			"		"		"	3		"		1	mA
		183			"	GND	5.5 V	"		"	"		"			"		"		"	14	01	"		"	"
		184			"	5.5 V	GND	"		"	"		"			"		"		"	6		"		"	"
		185			"	5.5 V	GND	"		"	"		"			"		"		"	11		"		"	"
	I_{IH3}	186	GND		2.4 V			"		"			"	5.5 V	GND	"					3		"		40	µA
		187	"		5.5 V			2.4 V		"			"	"	"	"					6	02	"		"	"
		188	"		5.5 V			5.5 V		"			2.4 V	"	"	"					11		"		"	"
		189	"		5.5 V			5.5 V		"			5.5 V	"	"	2.4 V					14		"		"	"
		190	24 V		2.4 V			5.5 V		"			"	"	"	5.5 V				K2	3		"		"	"
		191	"		5.5 V			2.4 V		"			"	"	"	"				"	6	03	"		"	"
		192	"		"			5.5 V		"			2.4 V	"	"	"				"	11		"		"	"
		193	"		"			"		"			5.5 V	"	"	2.4 V				"	14		"		"	"
	I_{IH4}	194	GND		5.5 V			"		"			"	"	"	5.5 V					3		"		1	mA
		195	"		"			"		"			"	"	"	"					6	02	"		"	"
		196	"		"			"		"			"	"	"	"					11		"		"	"
		197	"		"			"		"			"	"	"	"					14		"		"	"
		198	24 V		"			"		"			"	"	"	"				K2	3		"		"	"
		199	"		"			"		"			"	"	"	"				"	6	03	"		"	"
		200	"		"			"		"			"	"	"	"				"	11		"		"	"
		201	"		"			"		"			"	"	"	"				"	14		"		"	"
	I_{IH5}	202			GND	2.4 V	5.5 V	"		"	5.5 V		"			GND		24 V		K1	4	01	"		80	µA
		203			5.5 V	5.5 V	2.4 V	GND		"	"		GND			5.5 V		24 V		K1	5		"		80	"
		204	GND		GND			"		"			"	5.5 V	2.4 V	GND					13	02	"		160	"
		205	24 V		"			"		"			"	5.5 V	2.4 V	"				K2	13	03	"		160	"
	I_{IH6}	206			"	5.5 V	5.5 V	5.5 V		"	5.5 V		5.5 V			"		24 V		K1	4	01	"		2	mA
		207			5.5 V	5.5 V	5.5 V	GND		"	5.5 V		GND			5.5 V		24 V		K1	5		"		2	"
		208	GND		GND			"		"			"	5.5 V	5.5 V	GND					13	02	"		4	"
		209	24 V		GND			"		"			"	5.5 V	5.5 V	GND				K2	13	03	"		4	"
	I_{IL1}	210			5.5 V	0.4 V	5.5 V	5.5 V		"	5.5 V		5.5 V			5.5 V		24 V		K1	4	01	"		-3.2	"
		211			"	5.5 V	0.4 V	"		"	5.5 V		"			"		24 V		K1	5		"		-3.2	"
	I_{IL2}	212	GND		"			"		"			"	5.5 V	0.4 V	"					13	02	"		-6.4	"
		213	24 V		"			"		"			"	5.5 V	0.4 V	"				K2	13	03	"		-6.4	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
2 T_A = 125°C	I_{IL3}	214			0.4 V	5.5 V	5.5 V	5.5 V		GND	5.5 V		5.5 V			5.5 V		24 V		K1	3		3		-1.6	mA
		215			5.5 V	"	"	5.5 V		"	"		"			0.4 V		"		"	14	01	"		"	"
		216			"	"	"	0.4 V		"	"		"			5.5 V		"		"	6		"		"	"
		217			"	"	"	5.5 V		"	"		0.4 V			"		"		"	11		"		"	"
		218	GND		0.4 V			"		"			5.5 V	5.5 V	5.5 V	"					3		"		"	"
		219	"		5.5 V			"		"			"	"	"	0.4 V					14	02	"		"	"
		220	"		"			0.4 V		"			"	"	"	5.5 V					6		"		"	"
		221	"		"			5.5 V		"			0.4 V	"	"	"					11		"		"	"
		222	24 V		0.4 V			"		"			5.5 V	"	"	"				K2	3		"		"	"
		223	"		5.5 V			"		"			"	"	"	0.4 V				"	14	03	"		"	"
		224	"		"			0.4 V		"			"	"	"	5.5 V				"	6		"		"	"
		225	"		"			5.5 V		"			0.4 V	"	"	5.5 V				"	11		"		"	"
	I_{CC1}	226		GND	5 V	5 V	5 V	5 V		"	5.5 V		5 V			5 V	GND	24 V		K1	9	01	"		22	"
	I_{CC2}	227	GND		5 V			5 V		"			"	5.5 V	5 V	5 V					12	02	"		25	"
	I_{CC3}	228	24 V	GND	5 V			5 V	GND	"		GND	"	5.5 V	5 V	5 V	GND			K2	12	03	"		10	"
	I_{CC4}	229		"	4.5 V	4.5 V	GND	GND	50 mA	"	5.5 V		"			4.5 V	GND	24 V		K1	9	01	"		70	"
		230		"	4.5 V	4.5 V	GND	5 V		"	5.5 V	50 mA	GND			"	"	"			9		"		70	"
	I_{CC5}	231	GND	50 mA	GND			5 V		"	24 V		5 V	5.5 V	GND	5 V		"			12		"		75	"
		232	"		5 V			GND	50 mA	"	"		5 V	"	"	"		"			"	02	"		"	"
		233	"		"			5 V		"	"	50 mA	GND	"	"	"		"			"		"		"	"
		234	"		"			"		"	"		5 V	"	"	GND	50 mA	"			"		"		"	"
	I_{CC6}	235	24 V	-50 mA	GND			"		"	6 V		"	"	"	5 V		6 V		K2	"		"		12	"
		236	"		5 V			GND	-50 mA	"	"		"	"	"	"		"		"	"	03	"		"	"
		237	"		"			5 V		"	"	-50 mA	GND	"	"	"		"		"	"		"		"	"
		238	"		"			"		"	"		5 V	"	"	GND	-50 mA	"		"	"		"		"	"
	I_{CC7}	239		GND	"	5 V	5 V	"		"	5.5 V		5 V			5 V	GND	24 V		K1	16	01	"		20	"
		240	24 V	GND	"			"	GND	"		GND	5 V	5.5 V	5 V	5 V	GND			K2	1	03	"		20	"
	I_{CC8}	241	50 mA	GND	GND	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			5 V	GND	24 V		K1	16	01	"		50	"
		242	50 mA	GND	5 V	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			GND	GND	24 V		K1	16		"		50	"
	I_{CC9}	243	24 V	-50 mA	GND			5 V		"	6 V		5 V	5.5 V	GND	5 V		6 V		K2	1		"		55	"
		244	"		5 V			GND	-50 mA	"	"		5 V	"	"	"		"		"	"	03	"		"	"
		245	"		"			5 V		"	"	-50 mA	GND	"	"	"		"		"	"		"		"	"
		246	"		"			5 V		"	"		5 V	"	"	GND	-50 mA	"		"	"		"		"	"
3 T_A = -55°C	V_{IC}	247		GND	-10 mA	4.5 V	4.5 V	4.5 V		GND	4.5 V		4.5 V			4.5 V	GND	24 V		K1	3		3		-1.7	V
		248		"	4.5 V	"	"	-10 mA		"	"		4.5 V			"	"	"		"	6		"		"	"
		249		"	"	"	"	4.5 V		"	"		-10 mA			"	"	"		"	11	01	"		"	"
		250		"	"	"	"	"		"	"		4.5 V			-10 mA	"	"		"	14		"		"	"
		251		"	"	-10 mA	"	"		"	"		"			4.5 V	"	"		"	4		"		"	"
		252		"	"	4.5 V	-10 mA	"		"	"		"			"	"	"		"	5		"		"	"
		253	GND		-10 mA	"		"		"	24 V		"	4.5 V	4.5 V	"		"			3		"		"	"
		254	"		4.5 V	"		-10 mA		"	"		"	"	"	"		"			6		"		"	"
		255	"		"	"		4.5 V		"	"		-10 mA	"	"	"		"			11	02	"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
3 T_A = -55°C	V_{IC}	256	GND		4.5 V	4.5 V		4.5 V		GND	24 V		4.5 V	4.5 V	4.5 V	-10 mA		24 V			14	02	3		-1.7	V
		257	GND		4.5 V	"		"		"	"		"	"	-10 mA	4.5 V		"			13		"		"	"
		258	24 V		-10 mA	"		"		"	"		"	"	4.5 V	"		"			3		"		"	"
		259	"		4.5 V	"		-10 mA		"	"		"	"	"	"		"			6		"		"	"
		260	"		"	"		4.5 V		"	"		-10 mA	"	"	"		"			11	03	"		"	"
		261	"		"	"		"		"	"		4.5 V	"	"	-10 mA		"			14		"		"	"
		262	"		"	"		"		"	"		4.5 V	"	-10 mA	4.5 V		"			13		"		"	"
	I_{OFF}	263	24 V	GND	GND	2 V	4.5 V	4.5 V		"	4.5 V					GND	GND	"		K1	1	01	"		500	µA
		264	"	"	2 V	GND	4.5 V	4.5 V		"	4.5 V					2 V	"	"		K1	1		"		"	"
		265	"	"	GND				GND	"	24 V	GND		4.5 V	2 V	GND	"	"		K2	16		"		"	"
		266	"	"	2 V				"	"	"	"		"	GND	2 V	"	"		"	16	03	"		"	"
		267	"	"				GND	"	"	"	"	GND	"	2 V		"	"		"	9		"		"	"
		268	"	"				2 V	"	"	"	"	2 V	"	GND		"	"		"	9		"		"	"
	V_{OH}	269	"	"	4.5 V	4.5 V	GND	2 V		"	4.5 V		4.5 V			4.5 V	"	"		K1	7		"	19		V
		270	"	"	"	"	2 V	GND		"	"		4.5 V			"	"	"		"	7	01	"	"		"
		271	"	"	"	"	GND	4.5 V		"	"		2 V			"	"	"		"	10		"	"		"
		272	"	"	"	"	2 V	4.5 V		"	"		GND			"	"	"		"	10		"	"		"
		273	GND		2 V					"	24 V			4.5 V	GND	"					2		"	"		"
		274	"		GND					"	"			"	2 V	"					2		"	"		"
		275	"		4.5 V					"	"			"	GND	2 V					15		"	"		"
		276	"		4.5 V					"	"			"	2 V	GND					15	02	"	"		"
		277	"					2 V		"	"		4.5 V	"	GND						7		"	"		"
		278	"					GND		"	"		4.5 V	"	2 V						7		"	"		"
		279	"					4.5 V		"	"		2 V	"	GND						10		"	"		"
		280	"					4.5 V		"	"		GND	"	2 V						10		"	"		"
	V_{CESAT1}	281	600 mA	GND	0.8 V	0.8 V				"	4.5 V					4.5 V	GND	15 V	15 V	K3	1	01	"		0.9	"
		282	600 mA	"	4.5 V	0.8 V				"	4.5 V					0.8 V	GND	15 V	15 V	K3	1		"		"	"
		283	24 V	GND				4.5 V	GND	"	600 mA	GND	0.8 V	4.5 V	0.8 V		"	24 V	24 V	K4	9		"		"	"
		284	"	"				0.8 V	"	"	600 mA	"	4.5 V	"	"		"	24 V	"	K4	9	03	"		"	"
		285	"	"	4.5 V				"	"	24 V	"		"	"	0.8 V	"	600 mA	"	K4	16		"		"	"
		286	"	"	0.8 V				"	"	24 V	"		"	"	4.5 V	"	600 mA	"	K4	16		"		"	"
	V_{CESAT2}	287					0.8 V	4.5 V		"	4.5 V	600 mA	0.8 V					15 V	15 V	K1	10	01	"		"	"
		288					0.8 V	0.8 V	600 mA	"	4.5 V	"	4.5 V					"	15 V	K1	7		"		"	"
		289	GND	600 mA	0.8 V				"	"	15 V			4.5 V	0.8 V	4.5 V		"	24 V		2		"		"	"
		290	"					0.8 V	600 mA	"	"			"	"			"	"		7	02	"		"	"
		291	"					4.5 V		"	"	600 mA	4.5 V	"	"			"	"		10		"		"	"
		292	"		4.5 V					"	"		0.8 V	"	"	0.8 V	600 mA	"	"		15		"		"	"
	V_{OC}	293	"	10 mA	"			4.5 V		"	GND		4.5 V	5.5 V	4.5 V	4.5 V		GND			2		"		1.5	"
		294	"		"			"	10 mA	"	"		"	"	"	"		"			7	02	"		"	"
		295	"		"			"		"	"	10 mA	"	"	"	"		"			10		"		"	"
		296	"		"			"		"	"		"	"	"	"	10 mA	"			15		"		"	"
	I_{OC}	297	"		"					"				"	0.8 V	0.8 V	50 mA	24 V			16		"		7	mA
		298	"	50 mA	0.8 V					"				"	"	4.5 V		24 V			16	02	"		"	"
		299	"					4.5 V		"	24 V	50 mA	0.8 V	"	"						9		"		"	"
		300	"					0.8 V	50 mA	"	24 V		4.5 V	"	"						9		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
3 T_A = -55°C	I_{IH1}	301			2.4 V	GND	5.5 V	5.5 V		GND	5.5 V		5.5 V			5.5 V		24 V		K1	3		3		40	µA
		302			5.5 V	GND	5.5 V	5.5 V		"	"		"			2.4 V		"		"	14	01	"		"	"
		303			"	5.5 V	GND	2.4 V		"	"		"			5.5 V		"		"	6		"		"	"
		304			"	5.5 V	GND	5.5 V		"	"		2.4 V			"		"		"	11		"		"	"
	I_{IH2}	305			"	GND	5.5 V	"		"	"		5.5 V			"		"		"	3		"		1	mA
		306			"	GND	5.5 V	"		"	"		"			"		"		"	14	01	"		"	"
		307			"	5.5 V	GND	"		"	"		"			"		"		"	6		"		"	"
		308			"	5.5 V	GND	"		"	"		"			"		"		"	11		"		"	"
	I_{IH3}	309	GND		2.4 V			"		"			"	5.5 V	GND	"					3		"		40	µA
		310	"		5.5 V			2.4 V		"			"	"	"	"					6	02	"		"	"
		311	"		5.5 V			5.5 V		"			2.4 V	"	"	"					11		"		"	"
		312	"		5.5 V			5.5 V		"			5.5 V	"	"	2.4 V					14		"		"	"
		313	24 V		2.4 V			5.5 V		"			"	"	"	5.5 V				K2	3		"		"	"
		314	"		5.5 V			2.4 V		"			"	"	"	"				"	6	03	"		"	"
		315	"		"			5.5 V		"			2.4 V	"	"	"				"	11		"		"	"
		316	"		"			"		"			5.5 V	"	"	2.4 V				"	14		"		"	"
	I_{IH4}	317	GND		"			"		"			"	"	"	5.5 V					3		"		1	mA
		318	"		"			"		"			"	"	"	"					6	02	"		"	"
		319	"		"			"		"			"	"	"	"					11		"		"	"
		320	"		"			"		"			"	"	"	"					14		"		"	"
		321	24 V		"			"		"			"	"	"	"				K2	3		"		"	"
		322	"		"			"		"			"	"	"	"				"	6	03	"		"	"
		323	"		"			"		"			"	"	"	"				"	11		"		"	"
		324	"		"			"		"			"	"	"	"				"	14		"		"	"
	I_{IH5}	325			GND	2.4 V	5.5 V	"		"	5.5 V		"			GND		24 V		K1	4	01	"		80	µA
		326			5.5 V	5.5 V	2.4 V	GND		"	"		GND			5.5 V		24 V		K1	5		"		80	"
		327	GND		GND			"		"			"	5.5 V	2.4 V	GND					13	02	"		160	"
		328	24 V		GND			"		"			"	5.5 V	2.4 V	"				K2	13	03	"		160	"
	I_{IH6}	329			GND	5.5 V	5.5 V	5.5 V		"	5.5 V		5.5 V			"		24 V		K1	4	01	"		2	mA
		330			5.5 V	5.5 V	5.5 V	GND		"	5.5 V		GND			5.5 V		24 V		K1	5		"		2	"
		331	GND		GND			"		"			"	5.5 V	5.5 V	GND					13	02	"		4	"
		332	24 V		GND			"		"			"	5.5 V	5.5 V	GND				K2	13	03	"		4	"
	I_{IL1}	333			5.5 V	0.4 V	5.5 V	5.5 V		"	5.5 V		5.5 V			5.5 V		24 V		K1	4	01	"		-3.2	"
		334			"	5.5 V	0.4 V	"		"	5.5 V		"			"		24 V		K1	5		"		-3.2	"
	I_{IL2}	335	GND		"			"		"			"	5.5 V	0.4 V	"					13	02	"		-6.4	"
		336	24 V		"			"		"			"	5.5 V	0.4 V	"				K2	13	03	"		-6.4	"
	I_{IL3}	337			0.4 V	5.5 V	5.5 V	5.5 V		"	5.5 V		5.5 V			5.5 V		24 V		K1	3		"		-1.6	mA
		338			5.5 V	"	"	5.5 V		"	"		"			0.4 V		"		"	14	01	"		"	"
		339			"	"	"	0.4 V		"	"		"			5.5 V		"		"	6		"		"	"
		340			"	"	"	5.5 V		"	"		0.4 V			"		"		"	11		"		"	"
		341	GND		0.4 V			"		"			5.5 V	5.5 V	5.5 V	"					3		"		"	"
		342	"		5.5 V			"		"			"	"	"	0.4 V					14	02	"		"	"
		343	"		"			0.4 V		"			"	"	"	5.5 V					6		"		"	"
		344	"		"			5.5 V		"			0.4 V	"	"	"					11		"		"	"
		345	24 V		0.4 V			"		"			5.5 V	"	"	"				K2	3		"		"	"
		346	"		5.5 V			"		"			"	"	"	0.4 V				"	14	03	"		"	"
		347	"		"			0.4 V		"			"	"	"	5.5 V				"	6		"		"	"
		348	"		"			5.5 V		"			0.4 V	"	"	5.5 V				"	11		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open)																	Relays energized	Measured terminal	Device	Test Figure	Limits		Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17					Min	Max	
3 T_A = -55°C	I_{CC1}	349		GND	5 V	5 V	5 V	5 V		GND	5.5 V		5 V			5 V	GND	24 V		K1	9	01	3		22	mA
	I_{CC2}	350	GND		5 V			5 V		"			"	5.5 V	5 V	5 V					12	02	"		25	"
	I_{CC3}	351	24 V	GND	5 V			5 V	GND	"		GND	"	5.5 V	5 V	5 V	GND			K2	12	03	"		10	"
	I_{CC4}	352		"	4.5 V	4.5 V	GND	GND	50 mA	"	5.5 V		"			4.5 V	GND	24 V		K1	9	01	"		70	"
		353		"	4.5 V	4.5 V	GND	5 V		"	5.5 V	50 mA	GND			"	"	"			9		"		70	"
	I_{CC5}	354	GND	50 mA	GND			5 V		"	24 V		5 V	5.5 V	GND	5 V		"			12		"		75	"
		355	"		5 V			GND	50 mA	"	"		5 V	"	"	"		"			"	02	"		"	"
		356	"		"			5 V		"	"	50 mA	GND	"	"	"		"			"		"		"	"
		357	"		"			"		"	"		5 V	"	"	GND	50 mA	"			"		"		"	"
	I_{CC6}	358	24 V	-50 mA	GND			"		"	6 V		"	"	"	5 V		6 V		K2	"		"		12	"
		359	"		5 V			GND	-50 mA	"	"		"	"	"	"		"		"	"	03	"		"	"
		360	"		"			5 V		"	"	-50 mA	GND	"	"	"		"		"	"		"		"	"
		361	"		"			"		"	"		5 V	"	"	GND	-50 mA	"		"	"		"		"	"
	I_{CC7}	362		GND	"	5 V	5 V	"		"	5.5 V		5 V			5 V	GND	24 V		K1	16	01	"		20	"
		363	24 V	GND	"			"	GND	"		GND	5 V	5.5 V	5 V	5 V	GND			K2	1	03	"		20	"
	I_{CC8}	364	50 mA	GND	GND	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			5 V	GND	24 V		K1	16	01	"		50	"
		365	50 mA	GND	5 V	GND	5.5 V	5.5 V		"	5.5 V		5.5 V			GND	GND	24 V		K1	16		"		50	"
	I_{CC9}	366	24 V	-50 mA	GND			5 V		"	6 V		5 V	5.5 V	GND	5 V		6 V		K2	1		"		55	"
		367	"		5 V			GND	-50 mA	"	"		5 V	"	"	"		"		"	"	03	"		"	"
		368	"		"			5 V		"	"	-50 mA	GND	"	"	"		"		"	"		"		"	"
		369	"		"			5 V		"	"		5 V	"	"	GND	-50 mA	"		"	"		"		"	"
9 T_A = 25°C	t_{PLH1}	370	B	GND	A	A	5 V	5 V		"	5 V		5 V		R1	5 V	GND	15 V			"	01	4		50	ns
		371	B	"	5 V	A	5 V	5 V		"	5 V		"		R1	A	"	15 V			"		"		"	"
		372	15 V	"	"	R1		A	GND	"	D	GND	"	5 V	A	5 V	"				9		"		"	"
		373	"	"	"	"		5 V	"	"	D	"	A	"	"	"	"				9	03	"		"	"
		374	"	"	A	"		"	"	"		"	5 V	"	"	"	"	E			16		"		"	"
		375	"	"	5 V	"		"	"	"		"	"	"	"	A	"	E			16		"		"	"
	t_{PLH2}	376		"	5 V	5 V	A	A	C	"	5 V		"		R1	5 V	GND	15 V			7	01	"		45	"
		377		"	5 V	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		378	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		379	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		380	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		381	"		"			"		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{PHL1}	382	B	GND	A	A	5 V	"		"	5 V		"		R1	5 V	GND	"			1	01	"		50	"
		383	B	"	5 V	A	5 V	"		"	5 V		"		R1	A	"	"			1		"		"	"
		384	15 V	"	"	R1		A	GND	"	D	GND	"	5 V	A	5 V	"				9		"		"	"
		385	"	"	"	"		5 V	"	"	D	"	A	"	"	"	"				9	03	"		"	"
		386	"	"	A	"		"	"	"		"	5 V	"	"	"	"	E			16		"		"	"
		387	"	"	5 V	"		"	"	"		"	5 V	"	"	A	"	E			16		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
9 T_A = 25°C	t_{PHL2}	388		GND	5 V	5 V	A	A	C	GND	5 V		5 V		R1	5 V	GND	15 V			7	01	4		30	ns
		389		GND	5 V	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		390	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		391	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		392	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		393	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{TLH1}	394	20 V	E	A	A	5 V	5 V		"	5 V		"		R1	5 V		20 V			2	01	"		30	"
		395	"		5 V	A				"	5 V		"		R1	A	F	"			15		"		30	"
	t_{TLH2}	396	"	F	A	R1		5 V		"	20 V		"	5 V	A	5 V					2		"		15	"
		397	"		5 V	"		A	G	"	"		"	"	"	"		"			7	03	"		"	"
		398	"		"	"		5 V		"	"	H	A	"	"	"		"			10		"		"	"
		399	"		"	"		5 V		"	"		5 V	"	"	A	J	"			15		"		"	"
	t_{TLH3}	400		GND	"	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	15 V			7	01	"		15	"
		401		GND	"	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		402	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		403	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		404	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		405	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{TLH1}	406	20 V	E	A	A	5 V	5 V		"	5 V		"		R1	5 V		20 V			2	01	"		25	"
		407	"		5 V	A				"	5 V		"		R1	A	F	"			15		"		25	"
	t_{TLH2}	408	"	F	A	R1		5 V		"	20 V		"	5 V	A	5 V					2		"		18	"
		409	"		5 V	"		A	G	"	"		"	"	"	"		"			7	03	"		"	"
		410	"		"	"		5 V		"	"	H	A	"	"	"		"			10		"		"	"
		411	"		"	"		5 V		"	"		5 V	"	"	A	J	"			15		"		"	"
	t_{THL3}	412		GND	"	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	15 V			7	01	"		15	"
		413		GND	"	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		414	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		415	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		416	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		417	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_S	418		GND	"	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	"			7	01	"		35	"
		419		GND	"	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		420	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		421	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		422	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		423	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	V_{OH}	424	"	B	A			5 V		"	24 V		"	"	GND	5 V		24 V			2		"		V_S	V
		425	"		5 V			A	C	"	"		"	"	"	"		"			7	02	"		-0.25	"
		426	"		"			5 V		"	"	D	A	"	"	"		"			10		"			"
		427	"		"			"		"	"		5 V	"	"	A	E	"			15		"			"
		428	24 V	GND	A	R1		"	GND	"		GND	"	"	"	5 V	GND	B			16		"			"
		429	"	"	5 V	"		"	"	"		"	"	"	"	A	"	B			16	03	"			"
		430	"	"	"	"		A	"	"	C	"	"	"	"	5 V	"				9		"			"
		431	"	"	"	"		5 V	"	"	C	"	A	"	"	5 V	"				9		"			"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
10 T_A = 125°C	t_{PLH1}	432	B	GND	A	A	5 V	5 V		"	5 V		5 V		R1	5 V	GND	15 V			1	01	4		90	ns
		433	B	"	5 V	A	5 V	5 V		"	5 V		"		R1	A	"	15 V			1		"		"	"
		434	15 V	"	"	R1		A	GND	"	D	GND	"	5 V	A	5 V	"				9	03	"		"	"
		435	"	"	"			5 V	"	"	D	"	A	"	"	"	"				9		"		"	"
		436	"	"	A			"	"	"		"	5 V	"	"	"	"	E			16		"		"	"
		437	"	"	5 V			"	"	"		"	"	"	"	A	"	E			16		"		"	"
	t_{PLH2}	438		"	5 V	5 V	A	A	C	"	5 V		"		R1	5 V	GND	15 V			7	01	"		65	"
		439		"	5 V	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		440	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2	02	"		"	"
		441	"		5 V			A	C	"	"		5 V	"	"	"		"			7		"		"	"
		442	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		443	"		"			"		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{PHL1}	444	B	GND	A	A	5 V	"		"	5 V		"		R1	5 V	GND	"			1	01	"		"	"
		445	B	"	5 V	A	5 V	"		"	5 V		"		R1	A	"	"			1		"		"	"
		446	15 V	"	"	R1		A	GND	"	D	GND	"	5 V	A	5 V	"				9	03	"		"	"
		447	"	"	"			5 V	"	"	D	"	A	"	"	"	"				9		"		"	"
		448	"	"	A			"	"	"		"	5 V	"	"	"	"	E			16		"		"	"
		449	"	"	5 V			"	"	"		"	5 V	"	"	A	"	E			16		"		"	"
	t_{PHL2}	450		GND	5 V	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	15 V			7	01	"		45	"
		451		GND	5 V	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		452	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2	02	"		"	"
		453	"		5 V			A	C	"	"		5 V	"	"	"		"			7		"		"	"
		454	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		455	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{TLH1}	456	20 V	E	A	A	5 V	5 V		"			"		R1	5 V		20 V			2	01	"		45	"
		457	"		5 V	A				"			"		R1	A	F	"			15		"		45	"
	t_{TLH2}	458	"	F	A	R1		5 V		"	20 V		"	5 V	A	5 V		"			2		"		20	"
		459	"		5 V	"		A	G	"	"		"	"	"	"		"			7	03	"		"	"
		460	"		"	"		5 V		"	"	H	A	"	"	"		"			10		"		"	"
		461	"		"	"		5 V		"	"		5 V	"	"	A	J	"			15		"		"	"
	t_{TLH3}	462		GND	"	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	15 V			7	01	"		20	"
		463		GND	"	5 V	A			"	5 V	D	A		R1	"	GND	"			10		"		"	"
		464	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2	02	"		"	"
		465	"		5 V			A	C	"	"		5 V	"	"	"		"			7		"		"	"
		466	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		467	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{THL1}	468	20 V	E	A	A	5 V	5 V		"	5 V		"		R1	5 V		20 V			2	01	"		27	"
		469	"		5 V	A				"	5 V		"		R1	A	F	"			15		"		27	"
	t_{THL2}	470	"	F	A	R1		5 V		"	20 V		"	5 V	A	5 V		"			2		"		20	"
		471	"		5 V			A	G	"	"		"	"	"	"		"			7	03	"		"	"
		472	"		"			5 V		"	"	H	A	"	"	"		"			10		"		"	"
		473	"		"			5 V		"	"		5 V	"	"	A	J	"			15		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
10 T_A = 125°C	t_{THL3}	474		GND	5 V	5 V	A	A	C	GND	5 V		5 V		R1	5 V	GND	15 V			7	01	4		20	ns
		475		GND	5 V	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		476	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		477	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		478	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		479	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_S	480		GND	"	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	"			7	01	"		55	"
		481		GND	"	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		482	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		483	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		484	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		485	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	V_{OH}	486	"	B	A			5 V	C	"	24 V		"	"	GND	5 V		24 V			2	02	"	V_S -0.25		V
		487	"		5 V			A		"	"	D	"	"	"	"		"			7		"			"
		488	"		"			5 V		"	"		A	"	"	"		"			10		"			"
		489	"		"			"		"	"		5 V	"	"	A	E	"			15		"			"
		490	24 V	GND	A	R1		"	GND	"		GND	"	"	"	5 V	GND	B			16	03	"			"
		491	"	"	5 V	"		"	"	"		"	"	"	"	A	"	B			16		"			"
		492	"	"	"	"		A	"	"	C	"	"	"	"	5 V	"				9		"			"
		493	"	"	"	"		5 V	"	"	C	"	A	"	"	5 V	"				9		"			"
11 T_A = -55°C	t_{PLH1}	494	B	GND	A	A	5 V	5 V		"	5 V		5 V		R1	5 V	GND	15 V			1	01	4		90	ns
		495	B	"	5 V	A	5 V			"	5 V	GND	"		R1	A	"	15 V			1		"		"	"
		496	15 V	"	"	R1		A	GND	"	D	"	"	5 V	A	5 V	"				9	03	"		"	"
		497	"	"	"	"		5 V	"	"	D	"	A	"	"	"	"				9		"		"	"
		498	"	"	A	"		"	"	"		"	5 V	"	"	"	"	E			16		"		"	"
		499	"	"	5 V	"		"	"	"		"	"	"	"	A	"	E			16		"		"	"
	t_{PLH2}	500		"	5 V	5 V	A	A	C	"	5 V		"		R1	5 V	GND	15 V			7	01	"		65	"
		501		"	5 V	5 V	A			"	5 V	D	A		R1	"	GND	"			10		"		"	"
		502	GND	B	A			5 V		"	15 V		A	5 V	A	"		"			2		"		"	"
		503	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		504	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		505	"		"			"		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{PHL1}	506	B	GND	A	A	5 V	"		"	5 V		"		R1	5 V	GND	"			1	01	"		"	"
		507	B	"	5 V	A	5 V	"		"	5 V	GND	"		R1	A	"	"			1		"		"	"
		508	15 V	"	"	R1		A	GND	"	D	"	"	5 V	A	5 V	"				9	03	"		"	"
		509	"	"	"			5 V	"	"	D	"	A	"	"	"	"				9		"		"	"
		510	"	"	A			"	"	"		"	5 V	"	"	"	"	E			16		"		"	"
		511	"	"	5 V			"	"	"		"	5 V	"	"	A	"	E			16		"		"	"
	t_{PHL2}	512		GND	5 V	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	15 V			7	01	8		45	"
		513		GND	5 V	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		514	GND	B	A			5 V		"	15 V		5 V	5 V	A	"		"			2		"		"	"
		515	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		516	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		517	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"

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TABLE III. Group A inspection – Continued.

Subgroup	Symbol	Test no.	Test conditions (undesignated terminals are open) 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	Relays energized	Measured terminal	Device	Test Figure	Limits Min	Limits Max	Unit
11 T_A = -55°C	t_{TLH1}	518	20 V	E	A	A	5 V	5 V		GND	5 V		5 V		R1	5 V		20 V			2	01	4		45	ns
		519	"		5 V	A				"	5 V		"		R1	A	F	"			15		"		45	"
	t_{TLH2}	520	"	F	A	R1		5 V		"	20 V		"	5 V	A	5 V		"			2		"		20	"
		521	"		5 V	"		A	G	"	"		"	"	"	"		"			7	03	"		"	"
		522	"		"	"		5 V		"	"	H	A	"	"	"		"			10		"		"	"
		523	"		"	"		5 V		"	"		5 V	"	"	A	J	"			15		"		"	"
	t_{TLH3}	524		GND	"	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	15 V			7	01	"		20	"
		525		GND	"	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		526	GND	B	A			5 V		"	15 V		5 V	5 V	A						2		"		"	"
		527	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		528	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		529	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_{THL1}	530	20 V	E	A	A	5 V	5 V		"	5 V		"		R1	5 V		20 V			2	01	"		27	"
		531	"		5 V	A				"	5 V		"		R1	A	F	"			15		"		27	"
	t_{THL2}	532	"	F	A	R1		5 V		"	20 V		"	5 V	A	5 V		"			2		"		20	"
		533	"		5 V			A	G	"	"		"	"	"	"		"			7	03	"		"	"
		534	"		"			5 V		"	"	H	A	"	"	"		"			10		"		"	"
		535	"		"			5 V		"	"		5 V	"	"	A	J	"			15		"		"	"
	t_{THL3}	536		GND	5 V	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	15 V			7	01	"		20	"
		537		GND	5 V	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		538	GND	B	A			5 V		"	15 V		5 V	5 V	A						2		"		"	"
		539	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		540	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		541	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	t_S	542		GND	"	5 V	A	A	C	"	5 V		5 V		R1	5 V	GND	"			7	01	"		55	"
		543		GND	"	5 V	A	5 V		"	5 V	D	A		R1	"	GND	"			10		"		"	"
		544	GND	B	A			5 V		"	15 V		5 V	5 V	A						2		"		"	"
		545	"		5 V			A	C	"	"		5 V	"	"	"		"			7	02	"		"	"
		546	"		"			5 V		"	"	D	A	"	"	"		"			10		"		"	"
		547	"		"			5 V		"	"		5 V	"	"	A	E	"			15		"		"	"
	V_{OH}	548	"	B	A			5 V		"	24 V		"	"	GND	5 V		24 V			2		"	V_S		V
		549	"		5 V			A	C	"	"		"	"	"	"		"			7	02	"	-0.25		"
		550	"		"			5 V		"	"	D	A	"	"	"		"			10		"			"
		551	"		"			"		"	"		5 V	"	"	A	E	"			15		"			"
		552	24 V	GND	A	R1		"	GND	"		GND	"	"	"	5 V	GND	B			16		"			"
		553	"	"	5 V	"		"	"	"		"	"	"	"	A	"	B			16	03	"			"
		554	"	"	"	"		A	"	"	C	"	"	"	"	5 V	"				9		"			"
		555	"	"	"	"		5 V	"	"	C	"	A	"	"	5 V	"				9		"			"

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

c. Special subgroup shall be added to group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits of subgroups 10 and 11 as specified in table III herein.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End point electrical parameters shall be as specified in table II herein.

Table IV. Group C end-point electrical parameters. T_A = 25°C

Test	Device types 01, 03 limits		Device type 02 limits		Delta		Unit
	Min	Max	Min	Max	Min	Max	
I_{OFF}	---	500	---	---	-50	50	µA
I_{OC}	---	---	---	700	-70	70	µA

4.5 Methods of inspection. Methods of inspection shall be specified and as follows.

4.5.1 Voltage and current. All voltage values given are referenced to the microcircuit ground terminals. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to acquiring activity in addition to notification of the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of MIL-STD-883, method 5003), corrective action and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for “JAN” marking.

j. Packaging requirements (see 5.1).

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43123-1199.

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6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:.

V_{IC}	Input clamp voltage.
I_{OFF}	Collector off-state current.
I_{IL}	Low level input current (with V_{IN} = 0.4)
I_{CCH}	High level supply current. This is the supply current with the output high.
I_{CCL}	Low level supply current. This is the supply current with the output low.
V_{OL}	Low level output voltage. This at rated load for the TTL gates.
I_{OH}	High level output current.
V_{OH}	High level output voltage, TTL gate.
I_{OS}	Short-circuit output current TTL gate.
V_{OC}	Output clamp diode forward voltage
I_{OC}	Output clamp current (one output on).
t_S	Storage time.
t_{PLH}	Propagation delay time (low to high level output transition).
t_{PHL}	Propagation delay time (high to low level output transition).
t_{TLH}	Transition time (low to high level output transition).
t_{THL}	Transition time (high to low level output transition).

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	55325
02	55326
03	55327

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army – CR
Navy - EC
Air Force - 11
NASA – NA
DLA – CC

Preparing activity:
DLA - CC

Project 5962-2039

Review activities:
Army – MI, SM
Navy – AS, CG, MC, SH, TD
Air Force – 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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